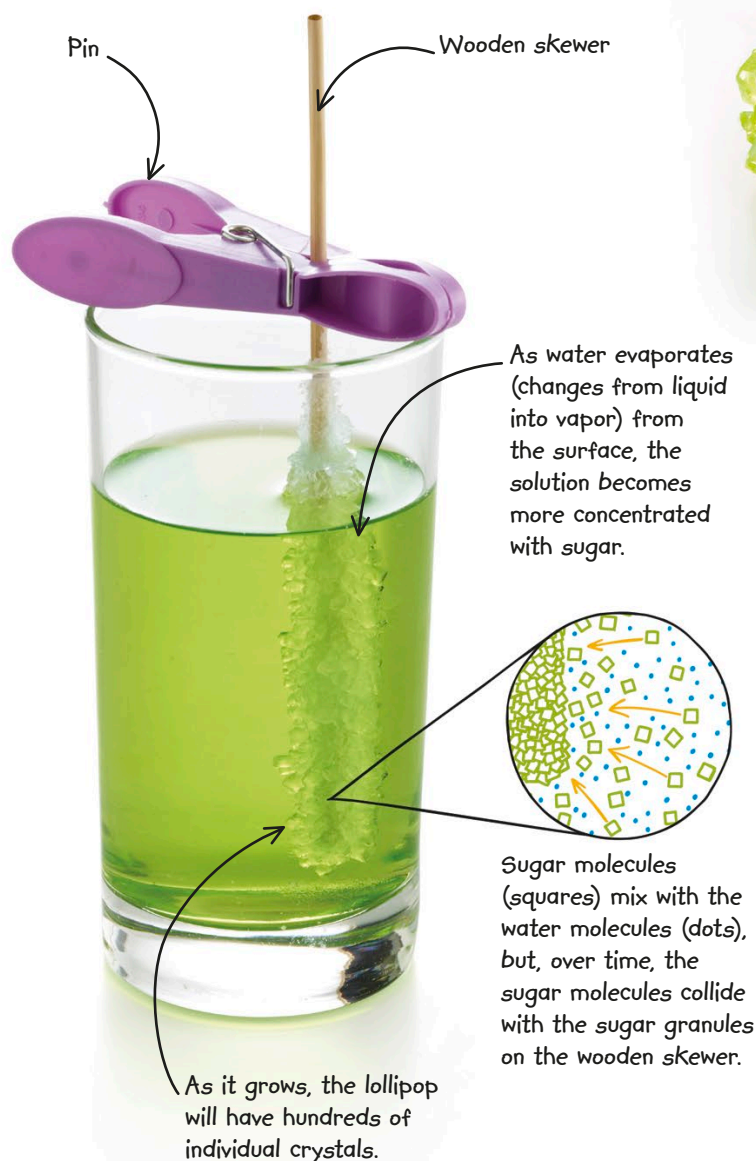


HOW IT WORKS

Each granule of sugar is a tiny crystal made of trillions of sugar molecules held together in a regular pattern. When you mix sugar with water, the sugar molecules break apart to mix among the water molecules, forming a solution. There is a high concentration of sugar molecules in the strong solution you made for this experiment. The concentration increases gradually, as the water evaporates from the surface of the solution. All of the molecules in the solution are slowly moving around, and when the sugar molecules collide with the sugar granules coating the skewer, they may stick. As more and more stick, the crystals grow—and so does your lollipop.



Sugar forms monoclinic crystals



CRYSTAL SHAPES

The way molecules join together determines the shape of the crystals. The shape of crystals that make up your lollipop is described as "monoclinic," where each crystal has three unequal sides.

REAL WORLD SCIENCE HOARFROST



In the same way that sugar molecules mixed with water molecules in this experiment, so too do water molecules mix with oxygen molecules in air. In cold weather, water molecules may join up, forming an icy coating called hoarfrost, which clings to surfaces.